Perplexity Improves While MMLU Lags: A Depth Study of Prune-then-Heal Language Models

Anonymous ACL submission

Abstract

Layer pruning followed by continued pretraining ("healing") is commonly judged by perplexity, but recovered next-token likelihood need not certify recovered capability. We study this distinction on OLMo-2 base models by retaining a pretrained prefix, appending a fresh tail, and tracking healing rather than only its endpoint. In the half-depth keep14 arm, perplexity improves throughout the observed budget while answer-letter MMLU and independent closed-book question answering remain far below the intact model. An unpruned 32-layer control continued on the same data stays near the base, separating the large gap from corpus shift alone. Scoring complete MMLU option text recovers more than scoring answer letters, revealing an interface component; however, a random-init model shares much of this content-score floor, so competence loss remains. Recovery is also policy dependent: an influence-selected ShortGPT endpoint preserves substantially more knowledge-sensitive performance than contiguous prefix truncation. Same-shape controls, paired trajectories, and a depth-wise readout probe delimit these conclusions without identifying a causal knowledge-storage layer. The resulting lesson is methodological: after structural intervention, likelihood, target capability, scoring interface, and recovery budget must be measured separately. Code and artifacts are available in our anonymous repository.

1 Introduction

Layer pruning is an attractive route to smaller language models: remove decoder blocks, continue pretraining the survivor, and recover much of its aggregate quality at a fraction of the original depth (Gromov et al., 2024; Men et al., 2024; Yang et al., 2024). This prune-then-heal recipe is often summarized by perplexity or a single benchmark average. Prior work has already shown that low perplexity and downstream quality need not rank pruned endpoints identically; what remains less characterized is how that mismatch evolves during healing and which controls distinguish structural loss from corpus shift or scoring-interface failure.

That gap motivates a trajectory-level measurement study. Perplexity averages local next-token events, whereas factual recall, answer selection, and task interfaces may depend differently on the computation removed by pruning. A lower perplexity after healing therefore does not by itself establish which evaluated behaviors have returned. Conversely, a low multiple-choice score can mix lost content competence with failure to map content onto an answer symbol.

Key observation. Under prefix truncation with a fresh tail, next-token fit and knowledge-sensitive evaluation recover at different observed rates. The gap persists beyond answer-letter MMLU on PopQA, TriviaQA, and NQ-open, but its magnitude depends on both the pruning policy and the scoring interface. We therefore treat recovery as a multi-axis object—likelihood, capability, interface, and training budget—rather than a scalar certified by perplexity.

We instantiate this study on OLMo-2-1124-7B (OLMo Team et al., 2025). Prefix arms retain k of 32 pretrained blocks, append two randomly initialized blocks, and heal on a fixed Dolmino stream. We compare their trajectories with the intact base, an unpruned model continued on the same stream, a non-contiguous ShortGPT endpoint, and same-shape frozen-front and fully random-init controls. Evaluation combines held-out likelihood, zero-shot likelihood scoring, two MMLU answer interfaces, and no-retrieval closed-book generation.

The evidence supports a bounded conclusion. The intact 32-layer control remains near the base after continued pretraining, while every tested half-depth point retains a substantial knowledge-sensitive gap. Complete-option scoring narrows the

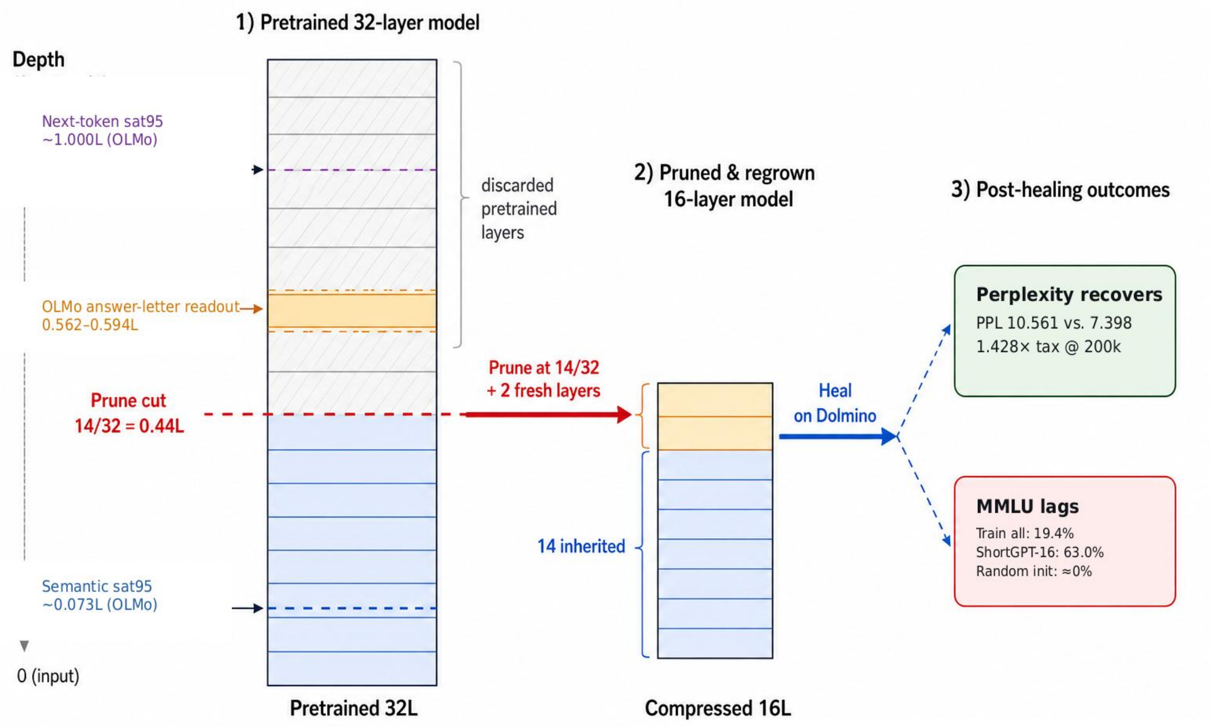

Figure 1: **Prune–regrow–heal and the measured recovery gap.** The main recipe retains the first $14/32$ pretrained blocks, appends two fresh blocks, and continues pretraining on Dolmino. On the same OLMo-2 base model, semantic probes become linearly readable early, MMLU answer letters become output-decodable only near 0.56–$0.59L$, and next-token top-1 agreement saturates at the top. These probe-dependent markers motivate the study but are not causal storage boundaries.

MMLU gap, so answer binding matters, but it does not remove the gap and has a large random-init fluency floor. ShortGPT performs markedly better than the prefix recipe, showing that the severity is policy dependent rather than a universal consequence of using 16 layers.

Contributions.

- We formulate post-pruning recovery as a multi-axis measurement problem and track likelihood and target capabilities along healing trajectories rather than judging only an endpoint.
- We provide continued-full-model, scoring-interface, policy, initialization, and adaptation controls, together with three independent closed-book tasks, to separate pruning effects from corpus shift and answer formatting.
- We relate the behavioral frontier to probe-dependent readout depths while explicitly limiting the result to a descriptive correlate, not a causal localization of knowledge.

2 Related Work

Table 1 positions our study relative to representative pruning pipelines. The distinction is one of analysis target: we do not propose another importance score, but ask what healing restores and which controls are required to establish it.

Layer removal and healing. ShortGPT and SLEB rank and remove redundant blocks (Men et al., 2024; Song et al., 2024); LaCo collapses adjacent layers (Yang et al., 2024). Other studies show that preferred layers depend on the target task and that simple depth pruning can be competitive (Siddiqui et al., 2024; Lu et al., 2024). Most directly related, Gromov et al. (2024) remove a contiguous block of deep layers and use continued pretraining to heal much of the perplexity penalty. We adopt this prune-then-heal setting as an object of study, while adding within-arm trajectories, an intact continued-pretraining control, and capability-specific evaluation.

Study family	Intervention	Adaptation	Reported recovery	Trajectory?	Key control(s)
ShortGPT / SLEB	ranked block removal	none	PPL + task endpoints	no	importance/policy variants
Gromov et al.	contiguous deep-block removal	continued pretraining	post-heal PPL + tasks	limited	depth and healing budget
Sheared / compact / DRPruning	structured pruning	CPT or distillation	compact-model endpoints	limited	data/training recipe
This study	prefix+fresh tail; ShortGPT endpoint	controlled healing	PPL + task + interface	**yes**	full32, init, freeze, policy

Table 1: **Positioning of the study.** This study does not introduce a new importance score. Its distinguishing object is the recovery trajectory, together with controls that separate structural intervention, continued-training drift, initialization/adaptation, pruning policy, and scoring interface. "Limited" means that intermediate training points are not the primary analysis object.

Structured prune-and-train. Sheared LLaMA and related pipelines combine structured pruning with continued pretraining, data allocation, or distillation to obtain compact endpoints (Xia et al., 2024; Muralidharan et al., 2024; Deng et al., 2025). SOLAR studies the inverse operation, duplicating blocks before continued training (Kim et al., 2024). Our two-block fresh tail instead provides a controlled regrowth interface after truncation; the paper evaluates recovery behavior, not a state-of-the-art compact model.

Proxy metrics beyond pruning. The distinction between optimization proxies and target behavior also appears in model merging, where parameter-space compatibility must ultimately be checked on the merged task suite (Yadav et al., 2023), and in continued pretraining, where corpus likelihood and downstream retention answer different questions. Our claim is therefore narrower than "perplexity is a bad metric": likelihood is essential for distributional fit, but after a structural or parameter intervention it should be paired with the capability the intervention is meant to preserve.

Knowledge and computation across depth. Feed-forward memories, causal tracing, and knowledge-neuron work associate factual behavior with middle-to-upper computation (Geva et al., 2021; Meng et al., 2022; Dai et al., 2022). Logit-lens and staged-inference analyses expose when information becomes readable through a particular output interface (nostalgebraist, 2020; Lad et al., 2024). Related depth-division work contrasts semantic and next-token readouts (Liu et al., 2026). We use these tools only to form and test a depth correlate. Neither a logit-lens transition nor pruning damage identifies where knowledge is stored.

3 Prune–Regrow–Heal Study

3.1 Intervention and model family

We study OLMo-2-1124-7B (OLMo Team et al., 2025), a 32-layer base language model without instruction tuning. For prefix arm `keep=` k, we retain the first k pretrained decoder blocks and append two randomly initialized blocks. We denote the cut location by $d_{\text{cut}} = k/32$ and resulting depth by $d_{\text{model}} = (k+2)/32$. Embeddings, final normalization, and the output head are copied exactly. The 1B OLMo-2 model provides a qualitative same-family replication.

The principal arm, keep14, is therefore a 16-layer model containing 14 inherited prefix blocks and two fresh tail blocks. We additionally evaluate an influence-selected ShortGPT-16 endpoint that inherits 16 non-contiguous blocks, including the original final block. It is a policy comparison, not a selection-only ablation, because inherited count and fresh-tail construction also differ.

3.2 Healing

All inherited prefix arms continue pretraining on the DCLM portion of `dolmino-mix-1124`, retokenized into 2,048-token windows. A disjoint 4096×2048 shard provides held-out likelihood. The maximum budget is 200k optimizer steps with effective batch size 128. In inherited runs, all trainable parameters use peak learning rate 2×10^{-5}; AdamW, cosine decay, mixed precision, and reconstruction checks are detailed in Appendix B.

Every depth arm completes the full 200k-step budget.

3.3 Control branches

Four controls delimit interpretation. The intact 32-layer model is continued on the same stream, separating pruning from corpus shift. Frozen-front keep14 trains the fresh tail and non-block modules without adapting inherited blocks. A fully random 16-layer model matches shape, data, batch, and maximum steps but uses a higher learning rate, so it is an operating point rather than an initialization-only ablation. Finally, ShortGPT tests policy dependence at the same resulting depth. Ongoing structure-isolation arms are not used in current claims.

3.4 Evaluation interfaces

Held-out perplexity is raw next-token loss with no chat template or inserted BOS. The principal downstream suite uses zero-shot likelihood scoring. Standard MMLU shows all four options and scores the single answer letter. A paired content protocol instead scores each complete option text, reporting both summed and length-normalized log-likelihood. Because a random-init model obtains a substantial normalized-content score, we interpret this protocol as an interface diagnostic rather than a direct knowledge measure.

We additionally evaluate PopQA (Mallen et al., 2023), TriviaQA (Joshi et al., 2017), and NQ-open (Kwiatkowski et al., 2019) with a shared zero-shot, no-retrieval `Question:`/`Answer:` generation protocol. The broader likelihood suite covers ten commonsense, reasoning, and comprehension tasks. Metrics, sample counts, prompts, checkpoint provenance, and strict architecture reconstruction appear in Appendix B.

OLMo-2-7B quantity	layer	fractional depth
semantic probe mean sat_{95}	2.33	$0.073L$
MMLU onset (chance+0.05)	18	$0.562L$
MMLU sat_{95}	19	$0.594L$
next-token sat_{95}	32	$1.000L$

Table 2: **Three ordered readout depths on the same OLMo model.** Semantic sat_{95} averages WiC, SST-2, and RTE depths measured relative to each final-layer probe score. MMLU uses the final norm and output head; next-token sat_{95} is exact top-1 agreement with the final layer. These are readout thresholds, not causal boundaries.

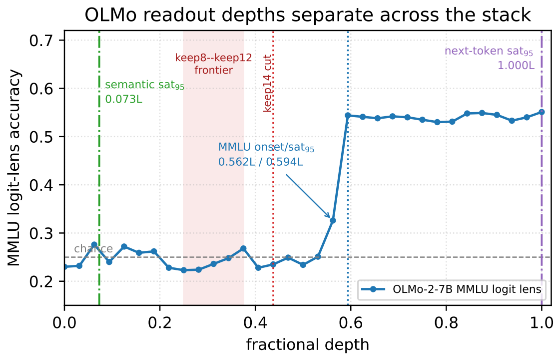

Figure 2: **Probe-dependent readout depths on OLMo-2-7B.** The shaded band marks prefix cuts used by the available behavioral frontier. Thresholds use different readouts and are descriptive; they are neither directly comparable causal boundaries nor locations of stored knowledge.

4 Depth-Dependent Correlates of Recovery

4.1 Likelihood and capability are different observables

Perplexity aggregates token-level negative log-likelihood over a corpus. A knowledge-sensitive task instead asks whether a particular answer can be selected or generated under a task interface. Healing can therefore improve the former without changing the latter at the same rate. This is not a claim that perplexity is uninformative; it is a claim that it is not a certificate for every behavior after a structural intervention.

4.2 Readability changes across depth

On the intact OLMo-2-7B model, a final-head logit lens keeps MMLU answer letters near chance through layer 17 and rises sharply at layers 18–19. Frozen semantic classifiers become readable much earlier, while exact next-token agreement with the final layer saturates only at the top. Figure 2 summarizes this probe-dependent ordering.

These probe-dependent readout depths are descriptive correlates, not causal boundaries. They motivate measuring recovery at multiple behavioral levels rather than assuming a single depth threshold, but they do not predict which capabilities will return after pruning.

4.3 Implications for interpreting recovery

The observations clarify three choices in the preceding study design. First, recovery is tracked along training because a single endpoint cannot distinguish delayed from absent change. Second, answer-letter and complete-option scoring are reported separately because they mix competence and readout differently. Third, pruning policy is varied because preserving selected upper blocks can change the result even at the same final depth. These are interpretation requirements, not a causal theory of where capability resides.

5 Experiments

5.1 Experimental setup

All main results use OLMo-2-7B under the base-model protocol in §3. Every depth arm completes the 200k-step budget. The central comparisons are the intact base, full32 continued-pretraining control, keep14+fresh2, ShortGPT-16, frozen-front, and fully random-init operating points. Detailed per-task results and all available checkpoints are in Appendix A.

5.2 Main results: likelihood recovers before knowledge-sensitive evaluation

Table 3 gives the central result. The half-depth keep14 arm reaches PPL 10.561 after 200k steps, yet remains far below the base on both MMLU interfaces and all three closed-book tasks. This is not explained by continued pretraining on Dolmino alone: the intact full32 control remains near the base in held-out likelihood, MMLU, and closed-book evaluation.

The MMLU interface matters but does not erase the result. Scoring complete option text raises keep14's above-chance recovery relative to answer-letter scoring, showing that content-to-symbol binding contributes to the standard metric. Yet fully random initialization attains nearly the same normalized-content score as frozen-front despite chance-level answer letters. The content protocol therefore contains a large fluency floor; neither interface alone is identified with stored knowledge.

The separation is not specific to multiple choice. At 200k, keep14 reaches only .142 PopQA answer containment, .294 TriviaQA exact match, and .060 NQ-open exact match, compared with .257, .636, and .205 for the intact base. These factual recall tasks use a shared no-retrieval generation interface, while full32 continued pretraining preserves much more of the base performance. Together with the two MMLU protocols, they rule out an answer-letter-only explanation.

Figure 4 makes the measurement issue visible. PPL and MMLU broadly co-vary across the strongest endpoints, but no scalar PPL threshold certifies capability: the random-init arm has lower PPL than frozen-front yet lower MMLU, while several structurally distinct endpoints cluster near chance. The plot therefore motivates reporting both axes; it is not evidence that they are globally uncorrelated.

5.3 Controlled comparisons

Same-shape initialization and adaptation. At 200k, keep14, frozen-front, and fully random init share the 16-layer shape. Their metric rankings disagree: frozen-front has worse PPL than random init but higher answer-letter MMLU, while train-all keep14 is strongest on both. Paired item-level tests

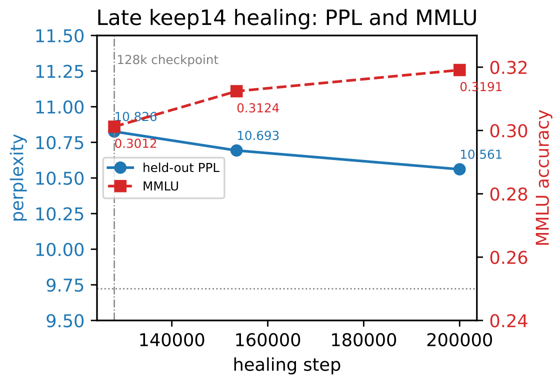

Figure 3: **Late keep14 healing continues on both axes, but the remaining capability gap is large.** From 128k to 200k, PPL improves from 10.826 to 10.561 and MMLU from .3012 to .3191; the intact base remains at .6053 MMLU. The figure supports slow late recovery, not a claim that MMLU is numerically constant.

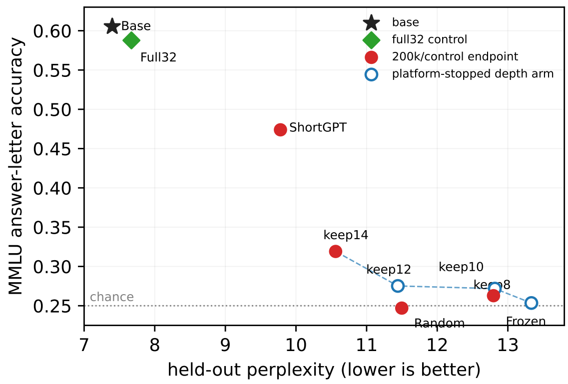

Figure 4: **Held-out PPL does not uniquely determine answer-letter MMLU across interventions.** All depth-arm endpoints are at 200k steps. Filled red points are control or ShortGPT endpoints; the dashed line connects the prefix ladder as a visual guide.

confirm the keep14 advantage over each control, but learning rate and trainable-module differences prevent an initialization-only or adaptation-only causal claim. Appendix Table 16 reports exact McNemar tests and paired-bootstrap intervals.

Full-model corpus control. Continuing the intact 32-layer model leaves PPL and both MMLU interfaces close to the original base and preserves most PopQA, TriviaQA, and NQ-open performance. Thus the large half-depth gap cannot be attributed to the healing corpus alone. A modest full32–base difference remains, so all recovery claims retain the continued-training control rather than treating the original base as an exact counterfactual.

Operating point	PPL↓	MMLU letter	MMLU content	PopQA	TriviaQA	NQ-open
Full base (32L)	7.398	.605	.471	.257	.636	.205
Full32 continued, 25k	7.670	.588	.466	.228	.572	.158
keep14 + fresh2, 200k	10.561	.318	.383	.142	.294	.060
ShortGPT-16, 200k	9.780	.474	.401	—	—	—
Frozen inherited front, 200k	12.797	.262	.360	.128	.248	.050
Fully random init, 200k	11.498	.247	.360	.111	.209	.063

Table 3: **Likelihood and knowledge-sensitive evaluation recover differently.** MMLU letter scores the answer symbol; MMLU content scores the length-normalized option text. PopQA reports normalized-answer containment; TriviaQA and NQ-open report exact match. Full32 continued pretraining stays near the base, whereas every tested half-depth operating point remains substantially below it. The elevated content score of fully random initialization exposes a fluency floor, so content accuracy is not interpreted as recovered knowledge.

Scoring-interface control. Across all nine evaluated arms, complete-option MMLU is less depth sensitive than answer-letter MMLU. ShortGPT remains stronger than the prefix arms under both interfaces, preserving the policy-dependence conclusion. The full table and protocol definition are in Appendix Table 17.

6 Analysis

6.1 Recovery varies across knowledge domains

Figure 5 separates absolute accuracy from recovery relative to each group's base margin. keep14 is strongest on the heterogeneous Other group (.377; 29.1% chance-adjusted recovery) and weakest on STEM (.294; 15.6%) and Humanities (.299; 16.2%). Social Science has higher raw accuracy (.335) but only 18.6% recovery because its base is much stronger. ShortGPT improves every group, most notably Other (.574) and Social Science (.554), yet still remains below full32.

Subject-level examples show that the pattern is not a single-domain artifact. Among high-sample subjects, keep14 retains relatively more on miscellaneous (n=783: .457 versus .803 base) and marketing (n=234: .440 versus .825), but little on professional law (n=1,534: .272 versus .460) and high-school statistics (n=216: .213 versus .514). These examples are descriptive: group membership mixes base difficulty, answer format, corpus exposure, factual recall, and reasoning. They do not establish that a particular knowledge type is stored in the removed blocks.

6.2 Prefix depth interacts with observed healing

keep8 and keep14 provide complementary within-arm trajectories. keep8's PPL falls smoothly, but a paired 45k–121k comparison finds no significant MMLU change. keep14 instead gains 1.79 MMLU points from 128k to 200k while PPL improves by 0.265 (Figure 3); the gain is statistically significant but leaves a large gap to the base. This is evidence that observed recovery rate depends on inherited depth within the prefix recipe.

All depth arms complete the full 200k-step budget.

6.3 What makes ShortGPT different?

ShortGPT and keep14 end at 16 blocks and complete the same 200k target budget, but differ along four coupled dimensions: ShortGPT inherits 16 rather than 14 pretrained blocks, selects them non-contiguously, retains original block 31, and uses no fresh tail. keep14 retains a contiguous prefix and appends two randomly initialized blocks. Embeddings, final normalization, and output heads are copied from the same base and therefore do not explain the difference.

The resulting endpoint gap is large: ShortGPT reaches .474 MMLU versus .319 for keep14 while also obtaining lower PPL. This establishes that nominal depth alone does not determine the attainable recovery frontier. It does *not* show that block 31 is uniquely responsible, that non-contiguous selection is optimal, or that fresh tails are harmful. The ongoing contiguous-16 and retained-final structure-isolation arms test these hypotheses separately.

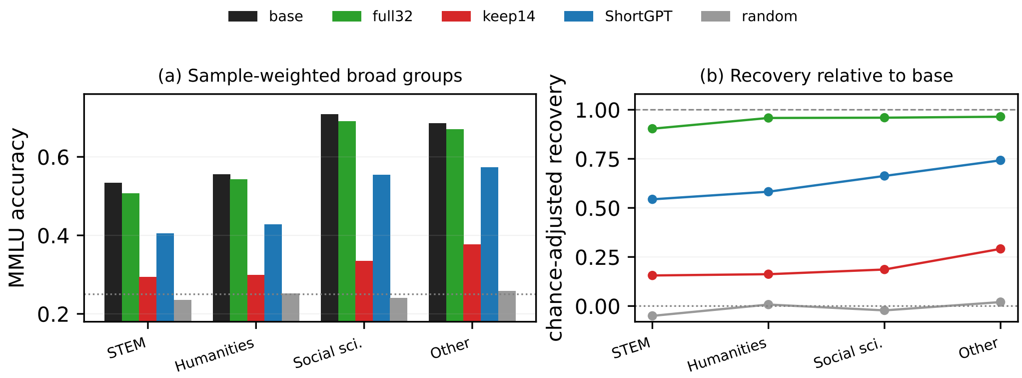

Figure 5: **MMLU recovery is heterogeneous across broad domains.** Accuracy is sample weighted over the standard four MMLU groups. Recovery is chance adjusted, $(a - 0.25)/(a_{\text{base}} - 0.25)$. Full32 remains close to the base in every group; ShortGPT consistently exceeds keep14, but neither half-depth arm closes the intact-model gap.

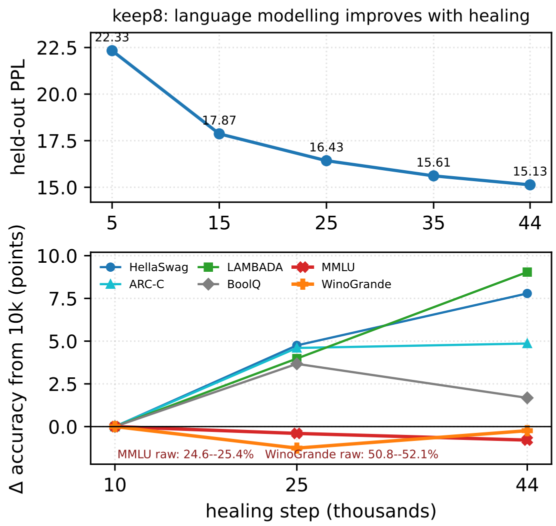

Figure 6: **The shallow keep8 arm improves in likelihood and ordinary tasks without detectable MMLU recovery over its observed trajectory.**

Arm	Budget	Inh./fresh	PPL↓	core6	MMLU
full base (32L)	—	32/0	7.398	.7037	.6053
keep8+fresh2 (10L)	200k	8/2	13.333	.5238	.2535
keep10+fresh2 (12L)	200k	12/2	11.443	.5669	.2752
keep14+fresh2 (16L)	200k	14/2	10.561	.5938	.3191
ShortGPT-16	200k	16/0	**9.780**	**.6215**	**.4739**

Table 4: **Depth and policy endpoints under a 200k target budget.** core6 averages six commonsense/reasoning tasks; Appendix Table 11 gives all cells. ShortGPT retains layers $[0\text{–}12, 16, 17, 31]$ and matches keep14 in total depth and completed 200k training, but differs in inherited count and fresh-tail construction.

6.4 Generality beyond the principal OLMo-2 setting

A more compressed OLMo-2-1B arm exhibits the same qualitative pattern of falling PPL with near-chance MMLU, providing same-family context rather than an independent replication. An existing Qwen3-8B endpoint supplies a limited cross-family check: a 14/36-layer prefix+fresh-tail model remains at .2495 MMLU after 200k steps, compared with .7297 for its base (Appendix Table 10). Qwen retains a smaller depth fraction than the OLMo keep14 recipe (39% versus 44%), which is one plausible contributor to its larger deficit, but the corpus, architecture, and missing content/closed-book controls prevent attribution. We therefore treat Qwen as directional evidence, not a matched replication.

7 Discussion

What does healing restore? The combined evidence argues against treating recovery as a single scalar. PPL measures how well the healed model fits held-out next-token events; it improves substantially in every inherited-prefix arm. That improvement is useful—the models recover fluent continuation and several commonsense tasks—but it does not uniquely determine answer selection or closed-book recall. The endpoint scatter makes the distinction concrete: models with similar PPL can occupy meaningfully different MMLU regimes, and random initialization can outrank a frozen inherited front in PPL while underperforming it on answer letters. We therefore interpret healing as partial recovery of distributional modeling plus a task-dependent remainder, not as a binary return of the original model.

Interface loss is real but insufficient. Complete-option MMLU recovers more smoothly than answer-letter MMLU, showing that part of the measured deficit lies in mapping content to the required output symbol. Yet the random-init fluency floor and the independent PopQA, TriviaQA, and NQ-open gaps rule out a formatting-only account. This distinction matters for evaluation: a single multiple-choice protocol can overstate damage when its interface is brittle, whereas a content-likelihood protocol can overstate recovery when generic fluency makes plausible options easy to score. Agreement across interfaces is more informative than designating either as ground truth.

Policy changes the attainable frontier. The step-matched keep14–ShortGPT comparison shows that half depth is not one operating point. Preserving selected upper blocks yields a much stronger likelihood–capability endpoint than retaining only a prefix and regrowing a tail. The result does not yet identify whether the advantage comes from the original final block, two additional inherited blocks, non-contiguous selection, or avoiding fresh layers. It does establish that conclusions drawn from one prefix recipe should not be generalized to layer pruning as a whole; the ongoing structure-isolation arms test these alternatives directly.

Which behaviors remain difficult? The residual deficit spans factual recall (PopQA, TriviaQA, and NQ-open), broad academic knowledge (all four MMLU groups), and some answer-interface behavior. In contrast, several continuation-style commonsense tasks recover more readily. This taxonomy is behavioral rather than anatomical. Knowledge-neuron and causal tracing studies motivate asking whether particular feed-forward computations support factual recall (Dai et al., 2022; Meng et al., 2022), but neither our subject groups nor ShortGPT's retention of block 31 identifies the missing substrate. Structure-isolation and causal interventions are required before making that claim.

A reporting protocol for structural compression. For prune-then-heal studies, we recommend reporting likelihood, target capability, recovery compute, and exact structural policy together. An intact model continued on the same stream controls for corpus drift, while same-shape initialization/adaptation controls expose fluency floors and inherited-computation effects.

Before claiming recovery from PPL alone: (1) evaluate an independent target capability; (2) test a second scoring or generation interface; (3) include a continued-full-model corpus control; and (4) report the exact structure and healing budget.

This checklist does not replace efficiency or endpoint-quality comparisons. It prevents a low PPL alone from being interpreted as evidence that the behaviors of interest have returned.

8 Conclusion

Prune-then-heal recovery is not one-dimensional. In the observed OLMo-2 prefix recipe, next-token likelihood improves well before knowledge-sensitive evaluation returns to the intact-model level. Continued full-model training, complete-option scoring, independent closed-book tasks, and same-shape controls show that both structural damage and answer interface contribute, while pruning policy materially changes the outcome. Depth-wise readouts provide a compatible but non-causal correlate. The practical conclusion is simple: after pruning, perplexity, capability, interface, and recovery budget should be reported as separate axes rather than collapsed into a single claim of healing.

Limitations

Within prefix-prune + fresh-tail healing, likelihood and knowledge-sensitive evaluation change at different observed rates. The intact full32 control makes corpus shift alone an insufficient explanation, and three closed-book datasets show that the result is not unique to answer-letter MMLU. However, this is a measurement study, not a new state-of-the-art pruning recipe.

All depth arms complete 200k steps, but deeper models execute more FLOPs per step, so the endpoint comparison is not token-matched. ShortGPT demonstrates policy dependence, but differs from keep14 in inherited count, final-layer retention, and fresh-tail construction. Contiguous-16 and retained-final structure-isolation arms are still training; until they finish, we do not attribute ShortGPT's advantage to a specific block or selection score.

Complete-option MMLU narrows the answer-letter gap, establishing an interface component. Its high random-init score also reveals a fluency floor, so it cannot be read directly as knowledge recovery. Conversely, closed-book exact match is sensitive to generation and normalization. We therefore rely on agreement across interfaces and datasets rather than designate one metric as ground truth.

The random-init arm uses a higher learning rate, and frozen-front changes the set of trainable modules. They are informative operating points, not clean causal ablations of initialization or adaptation. Likewise, logit-lens and linear-probe thresholds are readout dependent. Their ordering correlates with the behavioral frontier but neither localizes factual storage nor proves that removed layers contained the missing capability.

We report one training run per completed arm and the original runs did not fix an explicit seed. Item-level intervals quantify evaluation uncertainty, not training-seed variance. The controlled study covers one model family, one healing mixture, and primarily 7B scale; the OLMo-2-1B arm is qualitative, and the Qwen3-8B result is only a protocol-incomplete endpoint diagnostic trained on a different corpus. Other architectures, pruning policies, data mixtures, and longer budgets may yield different recovery orders. All evaluated checkpoints are base models; instruction tuning may redistribute task interfaces and change both answer binding and post-pruning recovery.

Compression studies should report likelihood, target-task behavior, scoring interface, and recovery compute separately. Perplexity remains useful for model fit, but after structural intervention it is not by itself a certificate that a particular evaluated capability has returned.

Ethical Considerations

This work studies how model compression via layer pruning and continued pretraining affects evaluated capability; it does not add a safety mechanism and changes none of the underlying model's risks such as hallucination, biased generation, or unsafe completion. The observed PPL–MMLU dissociation carries a practical caution: a compressed model that looks healthy on aggregate language-modelling metrics may still perform poorly on a target benchmark, and deployments should evaluate the capabilities they intend to preserve rather than rely on perplexity alone.

We collect no new human-subject data and recruit no annotators. All experiments use previously released artifacts—the OLMo-2 base model, the Dolmino corpus, and standard public benchmarks—under their respective licenses. Training and large-scale evaluation consume energy. We reuse a common training recipe and report checkpoint step explicitly rather than presenting unequal-step results as compute matched. Any release should preserve the licenses and usage restrictions of the underlying model, corpus, and benchmarks.

References

Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. 2022. Knowledge neurons in pretrained transformers. In *Annual Meeting of the Association for Computational Linguistics (ACL)*.

Hexuan Deng, Wenxiang Jiao, Xuebo Liu, Jing Li, Min Zhang, and Zhaopeng Tu. 2025. DRPruning: Efficient large language model pruning through distributionally robust optimization. In *Annual Meeting of the Association for Computational Linguistics (ACL)*.

Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. 2021. Transformer feed-forward layers are key-value memories. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Andrey Gromov, Kushal Tirumala, Hassan Shapourian, Paolo Glorioso, and Daniel A. Roberts. 2024. The unreasonable ineffectiveness of the deeper layers. *arXiv preprint arXiv:2403.17887*.

Mandar Joshi, Eunsol Choi, Daniel Weld, and Luke Zettlemoyer. 2017. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In *Annual Meeting of the Association for Computational Linguistics (ACL)*.

Dahyun Kim, Chanjun Park, Sanghoon Kim, Wonsung Lee, Wonho Song, Yunsu Kim, Hyeonwoo Kim, Yungi Kim, Hyeonju Lee, Jihoo Kim, et al. 2024. SOLAR 10.7B: Scaling large language models with simple yet effective depth up-scaling. In *Conference of the North American Chapter of the Association for Computational Linguistics (NAACL): Industry Track*.

Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. 2019. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466.

Vedang Lad, Wes Gurnee, and Max Tegmark. 2024. The remarkable robustness of LLMs: Stages of inference? *arXiv preprint arXiv:2406.19384*.

Hanzuo Liu, Xuan Qi, Chunyu Liu, Haotian Zhong, Yulong Wang, Rayying, Key, Alex Lamb, and Mingyu Gao. 2026. Understanding is done early: A depth division of labor in large language models and its use for unbounded-context memory. *arXiv preprint arXiv:2607.28263*.

Yao Lu, Hao Cheng, Yujie Fang, Zeyu Wang, Jiaheng Wei, Dongwei Xu, Qi Xuan, Xiaoniu Yang, and Zhaowei Zhu. 2024. Reassessing layer pruning in LLMs: New insights and methods. *arXiv preprint arXiv:2411.15558*.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Annual Meeting of the Association for Computational Linguistics (ACL)*.

Xin Men, Mingyu Xu, Qingyu Zhang, Bingning Wang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. 2024. ShortGPT: Layers in large language models are more redundant than you expect. *arXiv preprint arXiv:2403.03853*.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Saurav Muralidharan, Sharath Turuvekere Sreenivas, Raviraj Joshi, Marcin Chochowski, Mostofa Patwary, Mohammad Shoeybi, Bryan Catanzaro, Jan Kautz, and Pavlo Molchanov. 2024. Compact language models via pruning and knowledge distillation. In *Advances in Neural Information Processing Systems (NeurIPS)*.

nostalgebraist. 2020. Interpreting GPT: The logit lens. LessWrong. https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens.

OLMo Team, Pete Walsh, Luca Soldaini, Dirk Groeneveld, et al. 2025. 2 OLMo 2 Furious. *arXiv preprint arXiv:2501.00656*.

Shoaib Ahmed Siddiqui, Xin Dong, Greg Heinrich, Thomas Breuel, Jan Kautz, David Krueger, and Pavlo Molchanov. 2024. A deeper look at depth pruning of LLMs. *arXiv preprint arXiv:2407.16286*.

Jiwon Song, Kyungseok Oh, Taesu Kim, Hyungjun Kim, Yulhwa Kim, and Jae-Joon Kim. 2024. SLEB: Streamlining LLMs through redundancy verification and elimination of transformer blocks. In *International Conference on Machine Learning (ICML)*.

Mengzhou Xia, Tianyu Gao, Zhiyuan Zeng, and Danqi Chen. 2024. Sheared LLaMA: Accelerating language model pre-training via structured pruning. In *International Conference on Learning Representations (ICLR)*.

Prateek Yadav, Derek Tam, Leshem Choshen, Colin A. Raffel, and Mohit Bansal. 2023. TIES-Merging: Resolving interference when merging models. In *Advances in Neural Information Processing Systems*.

Yifei Yang, Zouying Cao, and Hai Zhao. 2024. LaCo: Large language model pruning via layer collapse. In *Findings of the Association for Computational Linguistics: EMNLP*.

A Complete Results

The main paper emphasizes the observed-budget perplexity–MMLU dissociation. This appendix reports all consolidated checkpoints, including the qualitative 1B comparison, within-arm trajectories, and chance-adjusted summaries.

A.1 Held-out perplexity checkpoints

Table 5 distinguishes progressing checkpoints from true endpoints. The fully random-init arm is a random language model with the same 16-layer architecture: its decoder blocks, embedding, final norm, and output head are initialized without OLMo-2 weights. It uses a different learning rate, so it is a budgeted operating point rather than an initialization-only ablation. Table 6 reports both immediate damage and the healed non-contiguous endpoint. Its 16 retained layers are selected by cosine block influence on 128 Dolmino windows and include the original final layer.

A.2 Within-architecture training trajectories

The complete 11-task keep8 trajectory in Table 7 is the cleanest 7B test of uniform under-training. Across a $4.4\times$ increase in optimization steps, nine reported tasks improve by 1.7–9.0 accuracy points, while MMLU stays within 0.79 point of chance and WinoGrande varies by only 1.26 points near chance. This rejects a uniform-delay account over the observed interval, but not recovery after substantially more training.

The OLMo-2-0425-1B arm shows the same qualitative separation (Figure 7; Tables 8–9): PPL falls $17.619 \rightarrow 15.628$ and HellaSwag/ARC-Easy/PIQA improve, while MMLU is .2495/.2558/.2529/.2480. Because this is one more-compressed arm from the same model family with a weaker base, we treat it as qualitative context, not an independent replication.

A.3 Capability recovery and continued healing

Table 12 exposes why a single average benchmark score would be misleading. Fully random initialization nearly matches inherited keep14 on ARC-Easy, PIQA, and BoolQ, but inherited initialization retains more signal on WinoGrande, MMLU, and several other downstream tasks. The late-healing comparison in Table 14 shows no broad rollback through 200k: from 153.5k to 200k, PPL improves by 0.132 while MMLU gains only 0.67 points and most other metrics move by less than one point. Table 15 additionally exposes raw and continuation-length-normalized scoring for BoolQ, CommonsenseQA, and SocialIQA, whose options have unequal lengths; the main tables consistently use raw accuracy for these tasks. Tables 17 and 18 report the complete MMLU interface control and the zero-shot closed-book results used in the main analysis. Table 16 uses the retained per-example predictions to compare final keep14 with fully random init on the same examples. All five differences favor keep14 under both exact McNemar and paired-bootstrap criteria; frozen-front is not included because its per-example predictions were not retained.

The confidence intervals in Table 13 distinguish the available frontier statistically: keep8, early

Scale	Arm	step	inherited	total layers	PPL↓	tax↓
7B	full base	—	32	32	7.398	1.000×
	keep8+fresh2	5k	8	10	22.331	3.019×
	keep8+fresh2	15k	8	10	17.868	2.416×
	keep8+fresh2	25k	8	10	16.426	2.220×
	keep8+fresh2	35k	8	10	15.612	2.110×
	keep8+fresh2	44k	8	10	15.131	2.045×
	keep10+fresh2	10k	10	12	17.239	2.330×
	keep10+fresh2	83.5k†	10	12	12.816	1.732×
	keep12+fresh2	111.5k	12	14	11.566	1.563×
	keep12+fresh2	124k†	12	14	11.443	1.547×
	keep14+fresh2	128k	14	16	10.826	1.463×
	keep14+fresh2	153.5k	14	16	10.693	1.445×
	keep14+fresh2	200k	14	16	10.561	1.428×
	ShortGPT-16	200k	16	16	**9.780**	**1.322**×
	frozen-front control	200k	14	16	12.797	1.730×
	fully random-init control	200k	0	16	11.498	1.554×
1B	full base	—	16	16	10.642	1.000×
	keep7+fresh2	50k	7	9	17.619	1.656×
	keep7+fresh2	100k	7	9	16.161	1.519×
	keep7+fresh2	147k	7	9	15.628	1.469×

Table 5: **All held-out perplexity checkpoints used in this study.** Every point uses the same disjoint 4096×2048 Dolmino validation shard (8,384,512 target tokens) and token-weighted eight-shard merging. The frozen-front arm retains the same pretrained prefix but does not update it; the fully random-init arm has the same 16-layer shape but loads no pretrained weights. ShortGPT inherits non-contiguous layers $[0\text{–}12, 16, 17, 31]$ and appends no fresh blocks. †Stopped after auxiliary tasks and MMLU plateaued; PPL was still decreasing.

ShortGPT point	PPL	core6	MMLU
step 0	401.124	—	.2620
step 200k	**9.780**	**.6215**	**.4739**

Table 6: **Non-contiguous ShortGPT endpoint.** Cosine block influence selects 16 of 32 pretrained layers, $[0\text{–}12, 16, 17, 31]$, with no fresh tail. The unhealed model is initially degenerate; after the same 200k-step Dolmino healing budget it recovers 63.0% of base above-chance MMLU.

Task	metric	10k	25k	44k	$\Delta_{10\to44}$
MMLU	raw	.2542	.2502	.2463	-0.79
HellaSwag	norm.	.3915	.4390	.4694	$+7.79$
ARC-Challenge	norm.	.2654	.3114	.3140	$+4.86$
ARC-Easy	norm.	.5581	.5930	.6103	$+5.22$
PIQA	norm.	.6632	.6801	.6937	$+3.05$
WinoGrande	raw	.5209	.5083	.5185	-0.24
OpenBookQA	norm.	.3000	.3360	.3320	$+3.20$
LAMBADA	raw	.3429	.3827	.4333	$+9.04$
BoolQ	raw	.5713	.6080	.5881	$+1.68$
CommonsenseQA	raw	.3923	.4103	.4423	$+5.00$
SocialIQA	raw	.3767	.3874	.4002	$+2.35$

Table 7: **Complete 11-task keep8 downstream trajectory.** The final column is the accuracy-point change from 10k to 44k under the pre-specified metric shown in column two; "norm." denotes accuracy based on continuation-length-normalized candidate log-likelihood. MMLU and WinoGrande remain near their chance floors while most other tasks improve. The held-out PPL checkpoints use a separate grid (5/15/25/35/44k; Table 5).

Model	step	HS	ARC-C	ARC-E	PIQA	WinoG	OBQA	PPL
1B full base	—	.683	.422	.735	.757	.643	.400	10.642
keep7+fresh2	50k	.411	.289	.568	.671	.531	.318	17.619
keep7+fresh2	100k	.438	.300	.593	.687	.526	.326	16.161
keep7+fresh2	147k	.450	.311	.595	.692	.530	.326	15.628
keep7+fresh2	148.5k	.448	.294	.606	.695	.544	.330	—

Table 8: **OLMo-2-1B core-task trajectory.** We report `acc_norm` (continuation-length-normalized likelihood accuracy) except for WinoGrande, where raw and normalized accuracy coincide. Abbreviations: HS = HellaSwag; ARC-C/E = ARC-Challenge/Easy; WinoG = WinoGrande; OBQA = OpenBookQA; PPL = perplexity.

keep10, and fully random init include chance, while keep12 and keep14 are above it. The subject-group table below shows that recovery is heterogeneous rather than concentrated in a single aggregate score.

A.4 MMLU subject-group results

Table 19 quantifies the broad-domain pattern. The inherited keep14 model is strongest in social science and the heterogeneous "Other" group, whereas science, technology, engineering, and mathematics (STEM) and humanities remain closer to chance. Table 22, placed at the end of the appendix as two side-by-side panels, reports every one of the 57 subjects at the 153.5k and final 200k keep14 checkpoints.

Model	step	MMLU	LAMBADA	BoolQ	CSQA	SIQA
1B full base	—	.3820	.6480	.624	.553	.422
keep7+fresh2	50k	.2495	.3536	.618	.368	.384
keep7+fresh2	100k	.2558	.3740	.597	.380	.394
keep7+fresh2	147k	.2529	.4021	.584	.391	.384
keep7+fresh2	150k	.2480	.3969	.577	.387	.386

Table 9: **OLMo-2-1B knowledge/comprehension trajectory.** MMLU remains at the .25 chance floor across the complete measured trajectory, whereas LAMBADA and commonsense tasks improve. Raw accuracy is shown for these five tasks.

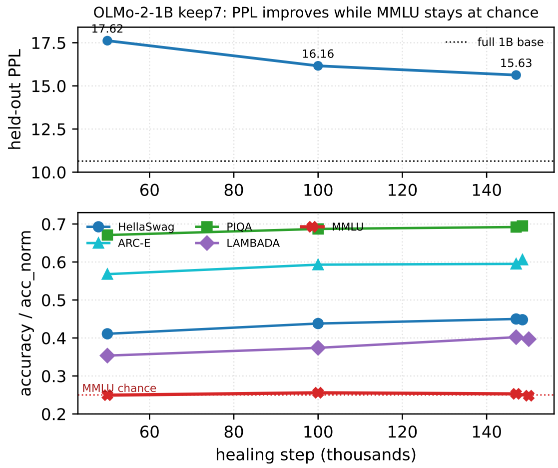

Figure 7: **Qualitative OLMo-2-1B trajectory.** PPL and several ordinary tasks improve while MMLU stays near chance over the observed keep7 interval.

B Evaluation and Reproducibility Details

B.1 Data, optimization, and architecture reconstruction

Healing uses the DataComp for Language Models (DCLM) portion of `dolmino-mix-1124`, tokenized into 2048-token windows with the OLMo-2 tokenizer. The training array is disjoint from the fixed validation shard of 4096 windows. All main arms use effective batch size 128 and a cosine schedule after 150 warmup steps. In the executed distributed runs, all parameters in inherited arms use peak/minimum learning rates $2\times10^{-5}/2\times10^{-6}$; the fully random-init arm uses $10^{-4}/10^{-5}$ for all parameters. We use AdamW with β_1=0.9, β_2=0.95, ϵ=10^{-8}, weight decay 0.1 on matrix parameters and zero on vectors, and global gradient-norm clipping at 1.0. Parameters and optimizer states use 32-bit floating point (fp32), while forward passes use bfloat16 (bf16) automatic mixed precision; gradient checkpointing is enabled. The maximum is 200k optimizer steps. No BA-

Qwen3-8B arm	Depth	PPL↓	MMLU letter
Full base	36/36	11.42	.7297
keep12 + fresh2, 200k	14/36	23.49	.2495

Table 10: **Cross-family endpoint diagnostic.** The Qwen3-8B arm heals on SlimPajama rather than Dolmino and is more compressed than the principal OLMo arm. It supports only the directional answer-letter finding; absolute PPL and recovery magnitude are not compared across families. Content-MMLU and closed-book controls are pending.

BILong or other synthetic task data is mixed into healing.

Runs do not set an explicit random seed, so the exact fresh initialization and initial shuffle are not reproducible from the command line alone. The keep14 run resumed once from step 34.5k with model, optimizer, and pseudorandom-number generator states restored. The checkpoint did not store the within-epoch data-loader offset, so the resumed iterator restarted that epoch's distributed shuffle; token counts in this paper are therefore nominal presentations rather than claims about unique examples.

For each inherited arm, launch-time checks require exact equality between the base and transplanted front blocks, embedding, final norm, and language-model (LM) vocabulary output head ($\max|\theta_{\mathrm{arm}} - \theta_{\mathrm{base}}| = 0$). Fresh-block norms are initialized to one and fresh query projections have standard deviation 0.02. Evaluation reconstructs the pruned architecture from checkpoint metadata and the saved parameter-state dictionary, then loads every tensor strictly. The keep14 models have 4.060B parameters.

B.2 Likelihood scoring and dataset sizes

For each multiple-choice candidate, the evaluator concatenates context and continuation, supplies the candidate tokens to the model, and sums their conditional log probabilities. This teacher-forced scoring never uses free generation. `acc_norm` divides this score by continuation character length, matching the evaluation convention used by the OLMo-2 baselines. WinoGrande uses its standard cloze format, in which each candidate fills the blank in the supplied sentence. MMLU candidates are answer letters under a fixed subject-description prompt. LAMBADA reports whether the entire final-word continuation is the greedy token sequence.

Evaluation integrity. Perplexity is merged as $\exp(\sum_g \mathrm{NLL}_g / \sum_g n_g)$ over eight shards, where NLL_g is shard g's summed negative log-likelihood and n_g is its token count; we never average per-shard perplexities. At keep14 steps 128k, 153.5k, and 200k, independent recomputation from all shard files reproduces every PPL and downstream cell to 10^{-9}. All evaluated task cells have zero NaNs and full sample counts. BoolQ contains the same two overlength passages in every run containing BoolQ; they are truncated consistently and still scored. No other task is truncated. The full 7B base reproduces expected OLMo-2 reference values (HellaSwag .805, MMLU .6053, LAMBADA .732), and the 1B base similarly reproduces its expected range.

C Mechanistic Probe Details

The MMLU logit lens uses $1{,}000$ MMLU test questions. At each layerwise hidden-state depth, the final-prompt-position hidden state is passed through the model's final normalization and output head; accuracy is the argmax among the letter tokens {A,B,C,D}. We define onset as the first layer exceeding chance by .05 and sat_{95} as the first layer reaching 95% of the top-layer above-chance signal. On OLMo these occur at layers 18 and 19, while sat_{99} is reached at layer 27. Complete OLMo and supplemental Qwen values appear in Table 21.

The new same-model OLMo probe adds frozen linear classifiers for WiC, SST-2, and RTE at every depth plus exact next-token top-1 agreement with the final layer. Their final-layer-relative sat_{95} depths are layers 1, 5, and 1, giving a mean of $2.33/32 = 0.073L$; because intermediate probes can outperform the final layer, this is a readout heuristic rather than a causal semantic boundary. Next-token agreement reaches sat_{95} only at layer 32 ($1.000L$). Together with MMLU sat_{95} at $0.594L$, this yields the same-model ordering semantic $<$ MMLU $<$ next-token. The keep14 step-200k evaluation is included throughout the main results and appendix.

Arm	step	MMLU	HS	ARC-C	ARC-E	PIQA	WinoG	OBQA	LAMB.	BoolQ	CSQA	SIQA
7B base full (32L)	—	**.6053**	.805	.571	.829	.811	.744	.462	.732	.815	.665	.502
keep8 (10L)†	44k	**.2463**	.469	.314	.610	.694	.519	.332	.433	.588	.442	.400
keep10 (12L)‡	10k	.2540	.443	.303	.585	.673	.524	.326	.404	.598	.421	.378
keep12 (14L)†	111.5k	**.2726**	.596	.407	.688	.736	.615	.376	.530	.610	.508	.415
keep14 (16L)	128k	**.3012**	.631	.426	.702	.747	.630	.402	.575	.639	.505	.423
keep14 (16L)	153.5k	**.3124**	.643	.442	.705	.745	.633	.406	.570	.606	.506	.441
keep14 (16L)	200k	**.3191**	.645	.438	.705	.745	.626	.404	.577	.638	.499	.434
ShortGPT-16	200k	**.4739**	.685	.476	.746	.758	.655	.408	.619	.729	.534	.442
frozen-front (16L)	200k	**.2628**	.595	.381	.666	.724	.646	.366	.513	.592	.453	.414
fully random-init (16L)	200k	**.2461**	.578	.414	.697	.733	.545	.384	.484	.614	.450	.416

Table 11: **Downstream zero-shot likelihood scoring (7B).** `acc_norm` is used for HS, ARC-C/E, PIQA, and OBQA; raw accuracy is used for MMLU, WinoG, LAMBADA, BoolQ, CSQA, and SIQA. Here `acc_norm` means accuracy from continuation-length-normalized log-likelihood. Appendix Table 15 reports both raw and normalized views for the final three tasks. Abbreviations: HS = HellaSwag; ARC-C/E = ARC-Challenge/Easy; WinoG = WinoGrande; OBQA = OpenBookQA; LAMB. = LAMBADA; CSQA = CommonsenseQA; SIQA = SocialIQA. The prefix ladder places keep8 at MMLU chance and final keep14 at 19.4% above-chance recovery. ShortGPT-16 reaches 63.0%, but also inherits two more pretrained blocks and the original final layer. Frozen-front recovers only 3.6%, while fully random init remains at chance despite nontrivial scores elsewhere and similar raw BoolQ accuracy to keep14 (.614 vs. .638). †Checkpoint not yet at the 200k budget. ‡Very early checkpoint.

Family	Task	chance	base	inherited	random init	recovery (%)
				reported score		*inherited / random*
surface	ARC-Easy	.25	.829	.705	.697	$78.6/77.2$
surface	PIQA	.50	.811	.745	.733	$78.9/74.9$
reasoning	HellaSwag	.25	.805	.645	.578	$71.1/59.1$
reasoning	ARC-Challenge	.25	.571	.438	.414	$58.5/51.1$
reasoning	WinoGrande	.50	.744	.626	.545	$51.6/18.4$
reasoning	OpenBookQA	.25	.462	.404	.384	$72.6/63.2$
reasoning	LAMBADA	.00	.732	.577	.484	$78.9/66.1$
reasoning	SocialIQA	.333	.502	.434	.416	$59.8/49.1$
reasoning	CommonsenseQA	.20	.665	.499	.450	$64.3/53.9$
in-context	BoolQ	.50	.815	.638	.614	$43.9/36.2$
answer-letter	**MMLU**	.25	.6053	.3191	.2461	$\mathbf{19.4/-1.1}$

Table 12: **Exact above-chance recovery for inherited keep14 and the fully random-init control at the matched step** 200**k.** Recovery is $100(x-c)/(b-c)$ for score x, chance c, and full-base score b. Family labels are descriptive: "surface" contains ARC-Easy and PIQA; "in-context" denotes passage-grounded BoolQ; "answer-letter" denotes the MMLU protocol; the remaining tasks are grouped descriptively as "reasoning." Reported scores use continuation-length-normalized likelihood for HellaSwag, ARC, PIQA, and OpenBookQA and raw accuracy otherwise. The negative random-init MMLU value means statistically chance-level performance. The random-init arm also randomizes lexical modules and uses a different learning rate, so the table is an operating-point comparison rather than an initialization-only or decoder-block-only ablation. Negative recovery denotes a score below the chance-adjusted reference.

Arm	MMLU	SE	approx. 95% CI
full 7B base	.6053	.0041	[.5972, .6134]
keep8, 44k	.2463	.0036	[.2392, .2534]
keep10, 10k	.2539	.0037	[.2467, .2611]
keep10, 83.5k	.2718	.0038	[.2644, .2792]
keep12, 111.5k	.2726	.0038	[.2652, .2800]
keep12, 124k	.2752	.0038	[.2678, .2826]
keep14, 128k	.3012	.0039	[.2936, .3088]
keep14, 153.5k	.3124	.0039	[.3047, .3201]
keep14, 200k	.3191	.0039	[.3114, .3268]
ShortGPT-16, 200k	.4739	.0042	[.4656, .4822]
frozen-front, 200k	.2628	.0037	[.2555, .2701]
fully random init, 200k	.2461	.0036	[.2390, .2532]

Table 13: **Marginal MMLU uncertainty across the 7B frontier.** Standard errors and Wald intervals use $n = 14{,}042$ questions. keep8, early keep10, and fully random init include the .25 chance floor; the latest keep10/keep12, frozen-front, all keep14 checkpoints, and ShortGPT are above it. These marginal intervals summarize each arm separately; Table 16 gives the stronger item-paired keep14–random comparison.

Metric	128k	153.5k	200k
held-out PPL	10.826	10.693	**10.561**
MMLU	.3012	.3124	**.3191**
HellaSwag	.631	.643	.645
ARC-Challenge	.426	.442	.438
ARC-Easy	.702	.705	.705
PIQA	.747	.745	.745
WinoGrande	.630	.633	.626
OpenBookQA	.402	.406	.404
LAMBADA	.575	.570	.577
BoolQ (raw acc.)	.639	.606	.638
CommonsenseQA (raw)	.505	.506	.499
SocialIQA (raw)	.423	.441	.434

Table 14: **Complete late keep14 trajectory.** From 153.5k to 200k, held-out PPL improves by 0.132 while MMLU gains only 0.67 points. Most other metrics remain within one point, with mixed signs; there is no broad capability rollback or late knowledge catch-up.

Task	metric	base	k8	k12	k14-p	k14-f	frozen	random
BoolQ	raw	.815	.588	.610	.606	.638	.592	.614
BoolQ	norm.	.753	.620	.657	.682	.689	.700	.619
CSQA	raw	.665	.442	.508	.506	.499	.453	.450
CSQA	norm.	.652	.405	.471	.479	.476	.467	.405
SIQA	raw	.502	.400	.415	.441	.434	.414	.416
SIQA	norm.	.547	.431	.460	.474	.474	.451	.443

Table 15: **Raw versus continuation-length-normalized accuracy.** The normalized score divides candidate log-likelihood by continuation character length before selecting an answer. The main results pre-specify raw accuracy for BoolQ, CommonsenseQA, and SocialIQA. Normalization can materially change BoolQ and SIQA, so both views are reported here. k14-p/k14-f denote keep14 at 153.5k/200k; frozen-front and random init are the other 200k controls.

Task	n	Δ (pp)	McNemar p	95% CI (pp)
MMLU	14,042	+7.30	5.96×10^{-49}	$[6.35, 8.26]$
LAMBADA	5,153	+9.35	1.84×10^{-39}	$[7.94, 10.75]$
CSQA	1,221	+4.83	2.97×10^{-4}	$[2.29, 7.37]$
BoolQ	3,270	+2.45	0.033	$[0.24, 4.65]$
SIQA	1,954	+1.84	0.042	$[0.15, 3.53]$

Table 16: **Paired keep14 versus fully random-init analysis at 200k.** Δ is keep14 minus random-init accuracy; the corresponding scores appear in Table 12. Items are matched by evaluation ID. McNemar uses an exact two-sided binomial test on discordant correctness pairs; 95% confidence intervals use 10,000 paired bootstrap resamples with seed 0. All intervals exclude zero and all $p < .05$. Frozen-front is excluded because per-example predictions were not retained.

Arm	letter	content raw	content norm.
Full base	.6054	.4400	.4706
Full32 continued	.5877	.4344	.4662
keep8	.2550	.3219	.3423
keep10	.2720	.3230	.3445
keep12	.2728	.3419	.3629
keep14	.3184	.3548	.3832
Frozen front	.2624	.3349	.3604
Random init	.2470	.3439	.3598
ShortGPT-16	.4742	.3764	.4012

Table 17: **MMLU scoring-interface control on all 14,042 items.** Letter scores one of A/B/C/D after displaying the options. Content scores the complete option text; the normalized variant is the pre-specified content headline.

Arm	PopQA contains	TriviaQA EM	NQ EM
Full base	.2571	.6355	.2050
Full32 continued	.2280	.5715	.1582
keep14	.1415	.2940	.0598
Frozen front	.1283	.2477	.0496
Random init	.1112	.2086	.0632
keep8†	.1246	.1577	.0368

Table 18: **Zero-shot, no-retrieval closed-book evaluation.** All arms use the same plain `Question:/Answer:` generation protocol. †keep8 has a different 10-layer shape and checkpoint budget and is included only as a depth reference.

Group	n	base	full32	keep14	ShortGPT	random
STEM	3,864	.534	.507	.294	.405	.236
Humanities	4,705	.555	.542	.299	.428	.252
Social science	3,077	.709	.690	.335	.554	.240
Other	2,396	.686	.670	.377	.574	.259

Table 19: **Sample-weighted MMLU broad-group accuracy.** full32 is the 25k continued-full-model control; keep14, ShortGPT, and random are their reported 200k endpoints. Group definitions follow the standard four-way MMLU taxonomy.

Task	split / mode	n	chance	metric
HellaSwag	validation, 4-choice	10,042	.25	acc_norm
ARC-Challenge	test, 4-choice	1,172	.25	acc_norm
ARC-Easy	test, 4-choice	2,376	.25	acc_norm
PIQA	validation, 2-choice	1,838	.50	acc_norm
WinoGrande	validation, 2-choice cloze	1,267	.50	accuracy
OpenBookQA	test, 4-choice	500	.25	acc_norm
MMLU	test, 57 subjects, 4-choice letters	14,042	.25	accuracy
LAMBADA-OpenAI	test, greedy final word	5,153	≈ 0	exact greedy accuracy
BoolQ	validation, yes/no	3,270	.50	accuracy
CommonsenseQA	validation, 5-choice	1,221	.20	accuracy
SocialIQA	validation, 3-choice	1,954	.333	accuracy

Table 20: **Evaluation suite and chance floors.** All tasks are zero-shot and likelihood scored; no free-form generation or chat template is used. `acc_norm` selects answers after dividing candidate log-likelihood by continuation character length. All listed examples are evaluated.

(a) OLMo-2-7B

L	frac.	acc.	gold LL	L	frac.	acc.	gold LL
0	0.0000	0.230	-14.499	17	0.5312	0.251	-11.947
1	0.0312	0.232	-14.024	18	0.5625	0.326	-11.652
2	0.0625	0.276	-13.244	19	0.5938	0.544	-8.517
3	0.0938	0.240	-12.939	20	0.6250	0.541	-8.038
4	0.1250	0.272	-13.112	21	0.6562	0.538	-7.002
5	0.1562	0.259	-12.793	22	0.6875	0.542	-5.694
6	0.1875	0.262	-12.206	23	0.7188	0.540	-6.038
7	0.2188	0.228	-11.715	24	0.7500	0.535	-4.847
8	0.2500	0.223	-11.116	25	0.7812	0.530	-5.091
9	0.2812	0.224	-11.294	26	0.8125	0.531	-5.701
10	0.3125	0.236	-11.493	27	0.8438	0.548	-5.345
11	0.3438	0.248	-12.051	28	0.8750	0.549	-5.456
12	0.3750	0.268	-11.847	29	0.9062	0.545	-5.402
13	0.4062	0.228	-11.499	30	0.9375	0.533	-5.338
14	0.4375	0.235	-11.295	31	0.9688	0.540	-4.475
15	0.4688	0.249	-11.555	32	1.0000	0.551	-3.791
16	0.5000	0.234	-11.960				

(b) Qwen3-8B

L	frac.	acc.	gold LL	L	frac.	acc.	gold LL
0	0.0000	0.272	-22.017	19	0.5278	0.237	-13.084
1	0.0278	0.234	-24.014	20	0.5556	0.237	-12.790
2	0.0556	0.240	-24.741	21	0.5833	0.236	-12.816
3	0.0833	0.236	-24.112	22	0.6111	0.235	-10.357
4	0.1111	0.233	-22.029	23	0.6389	0.232	-10.675
5	0.1389	0.237	-17.364	24	0.6667	0.236	-9.027
6	0.1667	0.242	-12.939	25	0.6944	0.621	-6.590
7	0.1944	0.234	-12.999	26	0.7222	0.625	-6.928
8	0.2222	0.229	-13.867	27	0.7500	0.624	-6.054
9	0.2500	0.241	-14.885	28	0.7778	0.635	-6.225
10	0.2778	0.237	-14.034	29	0.8056	0.572	-6.685
11	0.3056	0.229	-14.021	30	0.8333	0.629	-7.052
12	0.3333	0.228	-13.218	31	0.8611	0.630	-7.164
13	0.3611	0.225	-13.516	32	0.8889	0.629	-7.869
14	0.3889	0.229	-13.597	33	0.9167	0.629	-7.898
15	0.4167	0.231	-13.184	34	0.9444	0.638	-7.334
16	0.4444	0.237	-14.466	35	0.9722	0.636	-4.900
17	0.4722	0.236	-14.439	36	1.0000	0.632	-5.412
18	0.5000	0.231	-13.220				

Table 21: **Complete MMLU logit-lens trajectories.** Layer 0 is the embedding/input residual state and layer L follows decoder block L. Accuracy is four-letter MMLU accuracy; gold LL is full-vocabulary gold-letter log-likelihood on the same 1,000 questions. These are output-head readouts, not causal storage localizations.

Subject	n	base	k8	k12	k14-p	k14-f	frozen	rand
abstract_algebra	100	.310	.300	.240	.260	.270	.310	.240
anatomy	135	.622	.289	.296	.363	.363	.296	.215
astronomy	152	.730	.171	.289	.316	.322	.283	.178
business_ethics	100	.590	.250	.340	.390	.430	.260	.220
clinical_knowledge	265	.649	.242	.275	.317	.317	.257	.249
college_biology	144	.771	.278	.271	.347	.326	.250	.264
college_chemistry	100	.470	.190	.310	.270	.230	.260	.200
college_computer_science	100	.490	.240	.300	.300	.290	.240	.270
college_mathematics	100	.360	.200	.310	.240	.320	.270	.230
college_medicine	173	.566	.277	.341	.295	.289	.249	.214
college_physics	102	.422	.196	.304	.225	.275	.265	.147
computer_security	100	.680	.230	.250	.410	.400	.340	.290
conceptual_physics	235	.570	.238	.264	.311	.332	.243	.277
econometrics	114	.360	.246	.193	.246	.246	.325	.237
electrical_engineering	145	.600	.255	.276	.359	.366	.345	.248
elementary_mathematics	378	.397	.278	.254	.296	.272	.259	.222
formal_logic	126	.389	.302	.325	.198	.262	.183	.302
global_facts	100	.360	.340	.330	.280	.290	.320	.190
high_school_biology	310	.765	.223	.319	.319	.319	.239	.190
high_school_chemistry	203	.493	.261	.246	.256	.232	.202	.167
high_school_computer_science	100	.590	.290	.220	.300	.290	.280	.290
high_school_european_history	165	.752	.236	.236	.242	.309	.194	.206
high_school_geography	198	.783	.242	.308	.384	.328	.288	.207
high_school_government_and_politics	193	.870	.202	.285	.368	.358	.244	.197
high_school_macroeconomics	390	.636	.226	.338	.295	.321	.285	.221
high_school_mathematics	270	.296	.230	.278	.281	.248	.259	.219
high_school_microeconomics	238	.655	.239	.269	.277	.277	.252	.185
high_school_physics	151	.404	.238	.358	.225	.245	.298	.212
high_school_psychology	545	.807	.196	.297	.317	.358	.229	.207

Subject	n	base	k8	k12	k14-p	k14-f	frozen	rand
high_school_statistics	216	.514	.171	.338	.204	.213	.218	.194
high_school_us_history	204	.824	.314	.230	.319	.304	.299	.309
high_school_world_history	237	.797	.253	.270	.325	.354	.274	.253
human_aging	223	.650	.283	.139	.368	.359	.265	.341
human_sexuality	131	.756	.221	.298	.382	.382	.298	.321
international_law	121	.777	.248	.240	.281	.264	.355	.240
jurisprudence	108	.685	.204	.269	.287	.315	.231	.259
logical_fallacies	163	.693	.233	.215	.380	.350	.209	.221
machine_learning	112	.455	.170	.277	.375	.330	.277	.321
management	103	.777	.204	.359	.320	.369	.282	.194
marketing	234	.825	.286	.274	.440	.440	.286	.312
medical_genetics	100	.740	.170	.300	.370	.400	.340	.320
miscellaneous	783	.803	.271	.278	.436	.457	.268	.244
moral_disputes	346	.694	.225	.260	.329	.321	.272	.237
moral_scenarios	895	.279	.238	.238	.244	.253	.244	.238
nutrition	306	.693	.239	.307	.324	.327	.320	.242
philosophy	311	.711	.235	.270	.322	.318	.257	.222
prehistory	324	.741	.318	.275	.346	.358	.299	.238
professional_accounting	282	.415	.266	.245	.238	.241	.262	.280
professional_law	1534	.460	.244	.246	.259	.272	.245	.263
professional_medicine	272	.570	.199	.239	.228	.232	.301	.301
professional_psychology	612	.611	.243	.247	.320	.302	.260	.275
public_relations	110	.627	.282	.236	.291	.327	.309	.236
security_studies	245	.727	.367	.339	.294	.335	.245	.245
sociology	201	.836	.289	.279	.418	.428	.194	.294
us_foreign_policy	100	.850	.230	.340	.480	.450	.270	.340
virology	166	.524	.241	.313	.355	.386	.283	.313
world_religions	171	.836	.246	.292	.491	.509	.287	.322

Table 22: **Complete 57-subject MMLU results.** All values are accuracy. k14-p and k14-f are inherited keep14 at 153.5k and 200k; frozen-front and random init are the two other 200k controls. Subjects are split alphabetically across the two panels.